

FINAL (10:00-1:00)

Sun(2Nov)→ Network

Tues(4Nov)→ Software

Thurs(6 Nov)→ Microprocessor

Sun(9 Nov)→ Cyber Security

Tues(Phy)→ 11 Nov

Wed(BUS)→ 12 Nov

→ (only 2-5 PM)

University of Asia Pacific

Admit Card

Mid Semester Examination of Spring, 2025

Financial Clearance

PAID



Department of Computer Science and

	COURSE TITLE	CR.HR.	EXAM. SCHEDULE
	Communication and Computer Networks	3.00	
	Communication and Computer Networks Lab	1.50	
	Software Engineering	3.00	
4	CSE 314 Software Engineering lab	1.50	
5	CSE 315 Microprocessors and Microcontrollers	3.00	
6	CSE 316 Microprocessors and Microcontrollers Lab	1.50	
7	CSE 317 Computer and Cyber Security	3.00	
8	BUS 301 Business and Entrepreneurship	3.00	
9	PHY 301 Modern and Quantum Physics	3.00	

Total Credit: 22.50

1. Examinees are not allowed to enter the examination hall after 30 minutes of commencement of examination for mid semester examinations and 60 minutes for semester final examinations.

2. No examinees shall be allowed to submit their answer scripts before 50% of the allocated time of examination has elapsed.

3. No examinees would be allowed to go to washroom within the first 60 minutes of final examinations.

4. No student will be allowed to carry any books, bags, extra paper or cellular phone or objectionable items/incriminating paper in the examination hall. Violators will be subjects to disciplinary action.

This is a system generated Admit Card. No signature is required.

University of Asia Pacific
Department of Computer Science and Engineering
Final Examination, Spring 2025
Program: B.Sc. in CSE
3rd year 2nd Semester

Course Title: Data Communication and Computer Networks
Time: 3:00 hours

Credit Hour: 3.00

Course Code: CSE 311
Full Marks: 50

There are **Five** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

(see note)

- a. Briefly explain the core process of Network Address Translation (NAT). [4] CLO3/PLOa
Analyze the main challenges associated with NAT and their potential impact on network functionality.
- b. Apply the concept of IP addressing and routing to explain how packets are delivered across networks using Open Shortest Path First (OSPF) at the network layer and differentiate between EIGRP and RIP. [6] CLO3/PLOb

QUESTION 2 [10 MARKS]

- a. **Analyze** the DHCP process in a network where multiple devices are connected and show the difference between static routing and dynamic routing. [5] CLO2/PLOb
- b. Differentiate among the HTTP, SMTP, POP, and FTP protocols in terms of their connection-oriented or connectionless operation, transport layer protocol, and also consider stateful or stateless behavior. [5] CLO2/PLOb

OR

[10] CLO2/PLOb

Considering IP address is 172.16.20.5/20, answer the following:

- a) Subnet mask
- b) Number of subnets
- c) Block size / Hosts per subnet
- d) Valid hosts per subnet
- e) First and Last valid host
- f) Broadcast address
- g) Network address

QUESTION 3 [10 MARKS]

- a. Analyze the GSM network architecture in light of the security issues with SIM, IMEI, and HLR/VLR databases. [6] CLO4/PLOc
- b. Describe the handoff process and its potential impact on delay-sensitive applications when a mobile station is moving between two base stations. [4] CLO4/PLOc

QUESTION 4 [10 MARKS]

- a. Find the total bandwidth, carrier frequency (f_c), message signal (f_m) and modulation index (m), total modulation power (P_t) considering the given signal is $S(t) = 5\cos 1900\pi ft + 12\cos 2100\pi ft + 5\cos 2400\pi ft$. [5] CLO4/PLOc
- b. Host A wants to send 10 frames to host B. The hosts agreed to go with Selective Repeat ARQ. How many frames are transmitted by host A if every 6th frame that is transmitted by host A is either corrupted or lost? [5] CLO4/PLOc

QUESTION 5 [10 MARKS]

- a. Consider a sender wants to send 4 frames each of 8 bits, where the frames are 11001100, 10101010, 11110000 and 11000011. Using a checksum method to detect errors in transmitted data at the receiver. [5] CLO3/PLOb
- b. Find the carrier power and total sideband power of this AM signal considering the given signal is $S(t) = 20 [1 + 0.8 \cos (2\pi \times 10^3 t)] \cos (4\pi \times 10^6 t)$. [5] CLO3/PLOb

University of Asia Pacific
Department of Computer Science and Engineering
Final Examination, Spring 2025
Program: B.Sc. in CSE
3rd year 2nd Semester

Course Title: Software Engineering
Time: 3:00 hours

Credit Hour: 3.00

Course Code: CSE 313
Full Marks: 50

There are **Five** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

- a. Explain the desirability of high cohesion and low coupling in software design with an example. [5] CLO1/PLO(a)
- b. Explain the SOLID design principles in software engineering and the meaning of each letter "S, O, L, I and D". Provide a brief explanation and example for any two principles. [5] CLO1/PLO(a)

QUESTION 2 [10 MARKS]

"GlobalFinTech Inc." is undertaking the development of a sophisticated, multi-platform financial dashboard application intended to serve both web and mobile clients. The core application must be architected to handle significant complexity, featuring real-time data aggregation from disparate sources such as live stock market feeds, user portfolio databases, and third-party financial news APIs. This data requires substantial transformation, validation, and the application of complex business rules before it can be presented. The application's user interface needs to be highly dynamic and responsive, with live-updating charts, sortable data grids, and visual indicators that change based on real-time risk calculations. A critical requirement is ensuring the highest degree of testability and maintainability for the core financial logic, which must be isolated from the user interface to allow for thorough unit testing and to facilitate rapid, independent evolution of the UI based on user feedback. The development team is structured into specialized groups for backend core logic, UI/UX development, and quality assurance, necessitating a clean separation of concerns to enable effective collaboration.

The chief architect is evaluating two architectural patterns: the Model-View-Controller (MVC) pattern and the Model-View-ViewModel (MVVM) pattern.

- a. Compare how the MVC and MVVM patterns would manage the relationship between the user interface and the complex business logic in the context of GlobalFinTech's dashboard. Your comparison should explain the flow of data and the distinct responsibilities of each component within the two patterns. [6] CLO2/PLO(b)
- b. Select and justify the most suitable architectural pattern for this project by providing two specific reasons that directly address the challenges and requirements outlined in the scenario. *mvvm* [4] CLO2/PLO(b)

QUESTION 3 [10 MARKS]

- a. List the primary objectives of the Verification and Validation (V&V) process, and explain why full validation cannot be accomplished solely through inspections. [5] CLO2/PLO(b)
- b. Compare the principles of black-box, white-box, and gray-box testing techniques. Select the most effective technique and justify your answer? [5] CLO2/PLO(b)

QUESTION 4 [10 MARKS]

A national software company is developing a centralized hospital management system for a government health agency. The system will handle online patient registration, appointment scheduling, medical records, and payments, supporting up to 15,000 concurrent users. The estimated code size is 400 KLOC. [10] CLO3/PLO(k)

Project details:

- The system requires high reliability and medium product complexity.
- The development schedule must be strictly followed.
- Database size and virtual machine volatility should be moderate.
- The development team has excellent analyst capability but nominal application experience.
- The team has very low experience with virtual machines.

Calculate the effort (E) in man-months and development time (D) in months according to Table 1 and Table 2. Also calculate average staff size and their productivity.

Table 1: Project Type

Project	a	b	c	d
Organic	2.2	1.05	2.5	0.38
Semidetached	3.1	1.02	2.3	0.35
✓ Embedded	2.7	1.12	2.4	0.37

Table 2: Cost Drivers

Cost Drivers	Very Low	Low	Nominal	High	Very High
SCED	2.50	2.50	4.35	✓ 5.15	5.97
ACAP	2.75	3.88	4.00	5.15	✓ 6.87
AEXP	1.34	1.89	✓ 2.56	3.12	4.67
DATA	0.94	1.00	✓ 1.08	1.16	1.28
CPLX	2.90	3.30	✓ 3.50	4.95	5.56
RELY	2.20	2.92	✓ 3.41	3.78	4.65
VIRT	1.23	1.98	• 2.54	3.02	3.86
VEXP	✓ 2.78	3.14	✗ 4.25	4.98	5.69

QUESTION 5 [10 MARKS]

While using free public Wi-Fi at an airport lounge, you are uploading a confidential report to your company's cloud storage. The connection slows down, and a certificate warning appears, leading you to suspect a possible interception. [10] CLO4/PLO(f)

Interpret how a Man-in-the-Middle (MITM) attack works and recommend effective techniques to prevent such an attack in this scenario.

OR

A student portal's grade lookup accepts a `student_id` and builds SQL by concatenating input. An attacker submits `123' OR '1'='1` in the field and the query returns all grade records, giving them unauthorized access. [10] CLO4/PLO(f)

Demonstrate how this SQL injection attack works and recommend effective techniques to prevent such an attack in this scenario.

University of Asia Pacific
Department of Computer Science and Engineering
Final Examination, Spring 2025
Program: B.Sc. in CSE
3rd year 2st Semester

Course Title: Microprocessors and Microcontrollers
Time: 3:00 hours

Credit Hour: 3.00

Course Code: CSE 315
Full Marks: 50

There are **Five*** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

A company is designing two different types of embedded systems:

System A: A general-purpose computing device that requires large memory, a display unit, external storage, and frequent software updates. (Pro)

System B: A compact automatic temperature control unit that needs to read sensor data and control a motor in real time with minimal external components.

- a. **Determine** which system (A or B) should use a microprocessor and which should use a microcontroller. Give reasons for your answer. [4]
- b. **Describe** the block diagrams of the two processing units used in System A and System B. Give one example of each and compare them. [6]

QUESTION 2 [10 MARKS]

You are given the following problem statement:

Input three integer values x , y , and z (You may choose any suitable values of x and y as input, maintaining the given conditions) from the user, where $0 \leq x, y \leq 4$, $x > y$, and z can take the following values:

If $z = 0$, perform addition and display $x + y$

If $z = 1$, perform subtraction and display $x - y$

If $z = 2$, perform $x \wedge y$ and display the result

If $z = 3$, perform NOT x and display the result

Assume all flags are initially cleared (0) before execution.

- a. **Construct** an equivalent Assembly language program to implement the above logic. [6]
- b. For each instruction in your assembly program, **determine** which flags (CF, ZF, SF, OF, PF, AF) are affected and show their status (0 or 1) after execution. [4]

OR

You are given the following problem statement:

Input three integer values x, y, and z (You may choose any suitable values of x and y as input, maintaining the given conditions) from the user, where $0 \leq x \leq 3$, and y can take the following values:

If $y = 1$, print $x = x + 0h$

If $y = 2$, print $x = x + 2h$

If $y = 3$, print $x = x + 4h$

If $y = 4$, print $x = x - 1h$

Otherwise, print "invalid input"

Assume all flags are initially cleared (0) before execution

- a. **Construct** an equivalent Assembly language program to implement the above logic. [6]
- b. For each instruction in your assembly program, **determine** which flags (CF, ZF, SF, OF, PF, AF) are affected and show their status (0 or 1) after execution. [4]

QUESTION 3 [10 MARKS]

Identify the correct instruction, **execute** it to show changes in the register value and carry flag (CF). [10]

- (i) Shift $DI = 3A7FH$ right three places, with zeros moved into the leftmost bit.
- (ii) Move all bits in $AL = C3H$ left one place, making sure that a 0 moves into the rightmost bit position.
- (iii) Rotate all the bits of $AL = 5AH$ left in three places.
- (iv) Rotate carry right one place through $DX = B2C4H$.
- (v) Move the $DH = 76H$ register right one place, making sure that the sign of the result is the same as the sign of the original number.

QUESTION 4 [10 MARKS]

An embedded smart garden system uses a microcontroller to monitor and display soil moisture levels in real time. The system lights up LEDs to indicate soil conditions (dry, moderate, or moist) and displays the exact moisture percentage on a small screen.

- a. **Identify** the suitable microcontroller and display type for this system with justification. [4]
- b. **Analyze** whether LEDs, the display, or a combination of both provides the most effective feedback to the user in this system. Give reasons for your answer. [3]
- c. **Investigate** how improper timing or port setup can cause errors in the LED indicators and display, and suggest one method to ensure reliable system output. [3]

QUESTION 5 [10 MARKS]

A smart waste management system is planned for a modern city to monitor the level of waste in bins and ensure timely collection. The system should detect the fill level of each bin, identify if the lid is open or closed, and send alerts to the control room using wireless communication. The system should also record time and date for each update.

- a. **Identify** the suitable sensors, modules, and actuators for designing this system and justify their use. [3]
- b. **Illustrate** the block diagram and pseudo-code to analyze how the system works. [7]

University of Asia Pacific
Department of Computer Science and Engineering
Final Examination, Spring 2025
Program: B.Sc. in CSE
3rd year 2nd Semester

Course Title: Computer and Cyber Security
Time: 3:00 hours

Credit Hour: 3.00

Course Code: CSE 317
Full Marks: 50

There are **Five** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

- a. **Describe** the phases of ethical hacking and penetration testing. [5] CLO1/PLOa
- b. **Design** a packet filtering gateway (firewall) using these settings and policies on [5] CLO2/PLOa

Table 1 . You have to show the ACL table.

Policy	Firewall Setting
No outside Web access.	Drop all outgoing packets to any IP address, port 80.
No incoming TCP connections, except those for the institution's public Web server only.	Drop all incoming TCP SYN packets to any IP except 130.207.244.203, port 80.
Prevent streaming audio/video from eating up the available bandwidth.	Drop all incoming UDP packets — except DNS and router broadcasts.
Prevent your network from being used for a smurf DoS attack.	Drop all ICMP packets going to a “broadcast” address (130.207.255.255).
Prevent your network from being tracerouted.	Drop all outgoing ICMP TTL expired traffic.

Table1: Firewall policies and settings

QUESTION 2 [10 MARKS]

- a. **Describe** the steps for mitigating a DDoS attack. [5] CLO1/PLOa
- b. Malware and illegal access attempts often target your company's internal network. [5] CLO2/PLOa

As a network security engineer, **design** an effective IDS strategy to identify and prevent intrusions. Explain the detection approach and how your design aids in early detection and recovery from attacks.

QUESTION 3 [10 MARKS]

- a. **Analyze** if the given function can satisfy the properties of hash with some examples. [5] CLO3/PLOb

$$h(m) = 10 + (\sum(\text{ASCII value of each character}) \bmod 90)$$

$$\text{If } m = \text{"abc"}; \text{ then, } h(m) = 10 + (97+98+99) \% 90 = 10 + 24 = 34$$

[ascii value of 'a'=97]

- b. Describe how a VPN works, and **analyze** the security of browsing the internet with and without a VPN. [5] CLO3/PLOb

QUESTION 4 [10 MARKS]

- a. **Describe** DNS Cache Poisoning with appropriate figures. [5] CLO1/PLOa
- b. **Discuss** the steps to perform digital forensics in this case. Suppose a scammer calls a user pretending to be from Bkash customer care and asks for the OTP "to verify the account." After getting the OTP, the scammer withdraws all the money from the victim's account." [5] CLO1/PLOa

QUESTION 5 [10 MARKS]

$$p = 67, q = 41$$

- a. **Apply** RSA to Encrypt the message $m = \text{"UAP"}$ with the given values public_key $= \{n=2747, e=7\}$ [ascii value of 'A'=65] [5] CLO2/PLOa
- b. **Determine** the value of the private key (d) based on the given parameters on Question 5(a) [5] CLO2/PLOa

University of Asia Pacific
Department of Computer Science and Engineering
Final Examination, Spring 2025
Program: B.Sc. in CSE
3rd year 2nd Semester

Course Title: Modern and Quantum Physics

Course Code: PHY 301

Time: 3:00 hours

Credit Hour: 3.00

Full Marks: 50

There are **Five** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]

- a. Derive the equation of wavelength shift for Compton effect. [7]
- b. An incident γ -ray of wavelength $\lambda = 0.05 \text{ \AA}$ is scattered and the scattered wavelength is measured as $\lambda' = 0.0500024 \text{ \AA}$. Calculate the scattering angle θ . [3]

QUESTION 2 [10 MARKS]

Derive the time dependent Schrodinger's equation. [10]

OR

Derive the time independent Schrodinger's equation. [10]

QUESTION 3 [10 MARKS]

- a. Explain the wave-particle duality and the de Broglie hypothesis. [7]
- b. An electron has a measured de Broglie wavelength of 0.05 nm. Calculate (a) its momentum in kg·m/s and (b) its kinetic energy (in electronvolts). Assume non-relativistic motion. [3]

QUESTION 4 [10 MARKS]

- a. Explain the wave function. [4]
- b. Derive the mathematical form of Galilean transformation. Show that Newton's law remains invariant in this transformation. [6]

QUESTION 5 [10 MARKS]

- a. Describe angular momentum in quantum mechanics. [6]
- b. Show that $[\hat{x}, \hat{p}_x] = i\hbar$; where symbols have their usual meanings. [4]

University of Asia Pacific
Department of Computer Science and Engineering
Final Examination, Spring 2025
Program: B.Sc. in CSE
3rd Year 1st Semester

Course Title: Business and Entrepreneurship

Time: 3:00 hours

Credit Hour: 3.00

Course Code: BUS 301

Full Marks: 50

Part A

There are **three** Questions. Answer any **two** of them. Part marks are shown in the margins.

QUESTION 1 [15 MARKS]

A company is considering **three** projects (A, B, and C) that are **mutually inclusive**.

Year	Project A BDT	Project B BDT	Project C BDT
0 (Initial Investment)	480,000	720,000	600,000
1	200,000	240,000	280,000
2	200,000	260,000	300,000
3	200,000	280,000	320,000
4	200,000	320,000	360,000

Conditions & Constraints:

1. The total budget for selected projects must not exceed BDT 1,200,000.
2. The discount rate is **15%**.

Requirements:

- a. Calculate the **Payback Period** of all the projects. Select the project/s if the cutoff year is 3.5 years. [07]
- b. Calculate the **net present value (NPV)** of all the projects. Select the project/s based on NPV criteria. [08]

QUESTION 2 [15 MARKS]

SRK Company
Statement of Financial Position
As at 31 December 2023 and 2022

Assets	2023	2022
Current assets	BDT	BDT
Cash	841,400	898,800

Prepaid expenses	1,036,000	700,000
Accounts receivable	1,649,200	1,439,200
Inventory	1,764,000	1,617,000
Total current assets	5,290,600	4,655,000
Plant assets (net)	9,086,000	7,284,200
Total assets	<u>14,376,600</u>	<u>11,939,200</u>
Liabilities and Stockholders' Equity		
Current liabilities		
Accounts payable	2,240,000	2,030,000
Income taxes payable	609,000	588,000
Total current liabilities	2,849,000	2,623,600
Bonds payable	3,080,000	2,800,000
Total liabilities	5,929,000	5,423,600
Stockholders' equity		
Common stock (BDT 5 par)	4,060,000	4,200,000
Retained earnings	4,387,600	2,315,600
Total stockholders' equity	8,447,600	6,515,600
Total liabilities and stockholders' equity	<u>14,376,600</u>	<u>11,939,200</u>

*2021 Accounts receivable BDT 7,00,000.

*2021 Inventory BDT 8,00,000.

SRK Company

Statement of profit or loss and other comprehensive income For the year ended as at 31 December 2023 and 2022

Particulars	2023 (BDT)	2022 (BDT)
Revenue	1,260,000	1,050,000
Cost of the service	735,000	630,000
Gross Profit	525,000	420,000
Operating Expenses	245,000	210,000
Net Income	<u>280,000</u>	<u>210,000</u>

Requirements:

[5*3]

Calculate each ratio mentioned below for both years and explain the result of each ratio with the benchmark:

- The current ratio
- Receivable turnover ratio
- Profit margin ratio
- Gross profit rate
- Debt to total asset ratio

QUESTION 3 [15 MARKS]

ReBag collects discarded materials from garment factories (usually sold at low prices as scrap) and uses them to produce eco-friendly bags and accessories. Each product is handmade by local artisans and designed for the growing sustainable fashion market. Their products are branded as "Made in Bangladesh, Made for the Planet." In recent times social media (Instagram, Facebook, TikTok) is playing a big role in influencing purchase decisions. 65% of Gen Z buyers are willing to pay 10–15% more for sustainable products. [15]

The global upcycled fashion market is valued at USD 4.5 billion and growing at 7–8% per year. Bangladesh's RMG industry produces over 400,000 tons of textile waste annually, creating a huge opportunity for circular economy businesses. There are some firms that are in the same business like Aarong Earth & Jatra. The plan is to collect Textile and leather scraps from garment factories in Savar, Gazipur, and Narayanganj and use a small workshop in Mirpur with 15 local artisans in Year 1. Monthly production capacity is estimated at 2,000 to 3,000 units. ReBag will sell its product through own e-commerce website. Social media platforms (Facebook Shop, Instagram Store and Local lifestyle stores. Their Average selling price is BDT 1,200 per item and Average cost per item (materials, labor, overhead) is BDT 700. Their four-year projection is given below:

Year	Units Sold	Revenue (Million)	COGS (Million)	Gross Profit (Million)	Operating Expenses (Million)	Net Profit (Million)
2026 (Actual)	25,000	30	17.5	12.5	10	2.5
2027 (Projected)	45,000	54	30	24	14	10
2028 (Projected)	70,000	84	42	42	18	24
2029 (Projected)	100,000	120	60	60	25	35
2030 (Projected)	130,000	156	75	81	30	51

Until now, the business has collected an investment of BDT 8 million. Out of which they have collected 1 million through bootstrapping, 2 million through FnF's (they had to provide 10% of the business share in exchange) and 5 million through Angel Investor against 20% share. To increase their production capacity, they will need additional BDT 10 million. **Prepare a business pitch for "Bangladesh Investments (A venture Capital Firm)" to collect the necessary fund.** Use the information provided in the ReBag Bangladesh case to prepare your business pitch. If any required information is not given, you are allowed to use your imagination realistically to fill in the gaps. The pitch should include all the key chapters required for a business pitch presentation.

Part B

There are **four** Questions. Answer any **two** of them. Part marks are shown in the margins.

QUESTION 4 [10 MARKS]

A **sole proprietorship** often faces limitations in capital and human resources. To overcome these constraints, businesses may adopt a **partnership** or form a **company** that offers greater financial strength and shared expertise.

Requirements:

- a. Differentiate between a Partnership Business and a Company. [06]
- b. Explain the steps involved in establishing a company, from registration to commencement of operations. [04]

QUESTION 5 [10 MARKS]

GreenWave Electronics Ltd. produces an advanced eco-friendly smart thermostat. The following annual cost data and selling information are available:

Cost/Revenue Item	Amount
Factory rent	\$90,000
Components and materials	\$60,000
Salaries of administrative staff	\$150,000
Packaging and shipping	\$12,000
Direct labor	\$36,000
Marketing and advertising campaigns	\$50,000
Sales commission	\$8,000
Equipment depreciation	\$20,000
Selling price	\$200,000

Requirements:

- a. Define breakeven analysis and its importance. [02]
- b. Calculate the following: [08]
- Identify which of the above costs are fixed and which are variable.
 - Calculate the total fixed cost, total variable cost per unit, and selling price per unit if total production is 1,000 units.
 - Determine the break-even point in units and in sales dollars.
 - Compute the number of units that must be sold to achieve the target profit of \$120,000.

QUESTION 6 [10 MARKS]

- a. Discuss different pricing methods and their purposes. [06]
- b. Discuss different business stages and optimal funding sources in those stages. [04]

QUESTION 7 [10 MARKS]

For a company planning to enter the telecommunications industry, analyze and evaluate the following aspects to assess the industry's attractiveness and future potential. Do an Industry Trend Analysis using Porter's Five Forces and/or PEST/PESTLE Framework. [10]